

Digital Control System Analysis and Design

Chap. 1 Introduction

Chap. 2 Discrete Time Systems and the z-Transform

Chap. 3 Sampling and Reconstruction

Chap. 4 Open-Loop Discrete-Time Systems

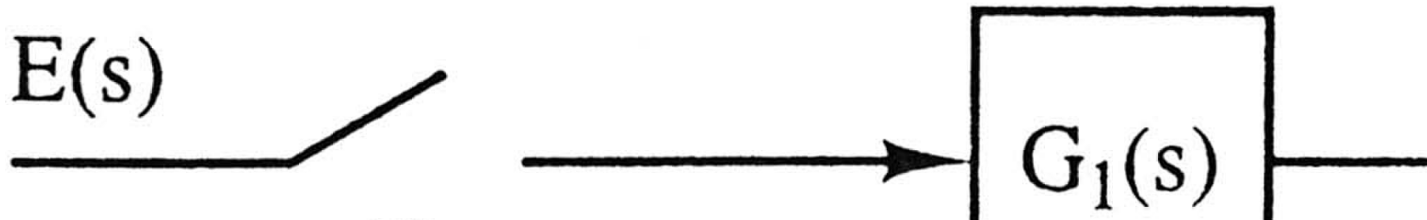
Chap. 5 Closed-Loop Systems

Chap. 6 System Time-Response Characteristics

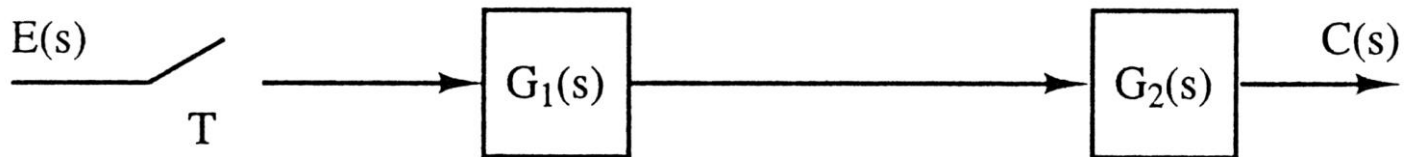
Chap. 7 Stability Analysis Techniques

Chap. 8 Digital Controller Design

5.2 Preliminary Concepts



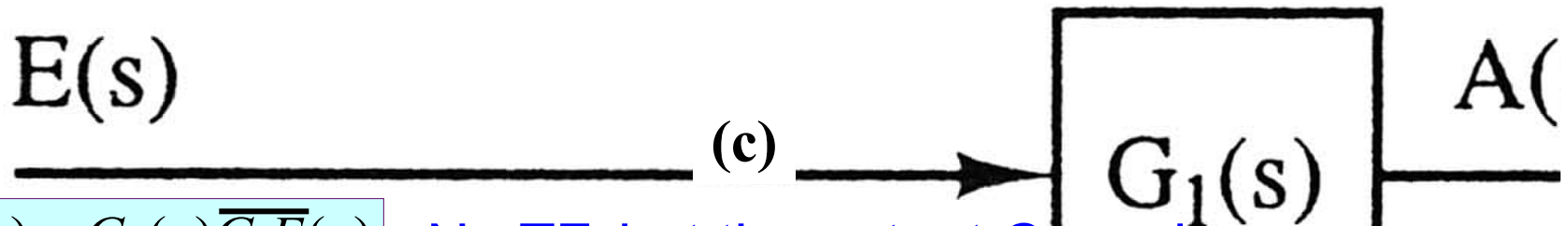
$$C(z) = G_1(z)G_2(z)E(z)$$



(b)

$$C(z) = \overline{G_1} \overline{G_2}(z)E(z)$$

Figure 5-1 Open-loop sampled-data systems.



$$C(z) = G_2(z) \overline{G_1} \overline{E}(z)$$

No TF, but the output C can be expressed as a function of the input E.

$E(s)$

$$C(z) = D(z)G(z)E(z)$$

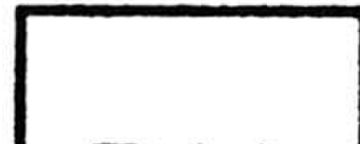


Figure 5-1 Open-loop sampled-data systems.

Compare (d) with (a).

Closed-Loop Systems

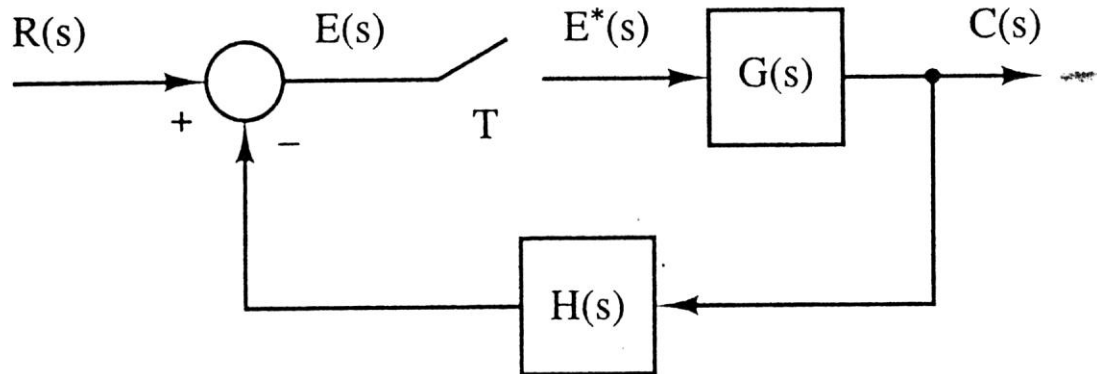


Figure 5-2 Closed-loop sampled-data system.

$$\begin{cases} C(s) = \underline{G(s)E^*(s)} & \text{(5-5)} \\ E(s) = R(s) - \underline{H(s)C(s)} \end{cases}$$

$$\rightarrow E(s) = R(s) - G(s)H(s)E^*(s)$$

$$E^*(s) = R^*(s) - \overline{GH}^*(s)E^*(s)$$

$$E^*(s) = \frac{R^*(s)}{1 + \overline{GH}^*(s)} \quad \text{(5-9)}$$

$$\Rightarrow \boxed{C(z) = \frac{G(z)R(z)}{1 + \overline{GH}(z)}} \quad \text{(5-11)}$$

Transfer function?

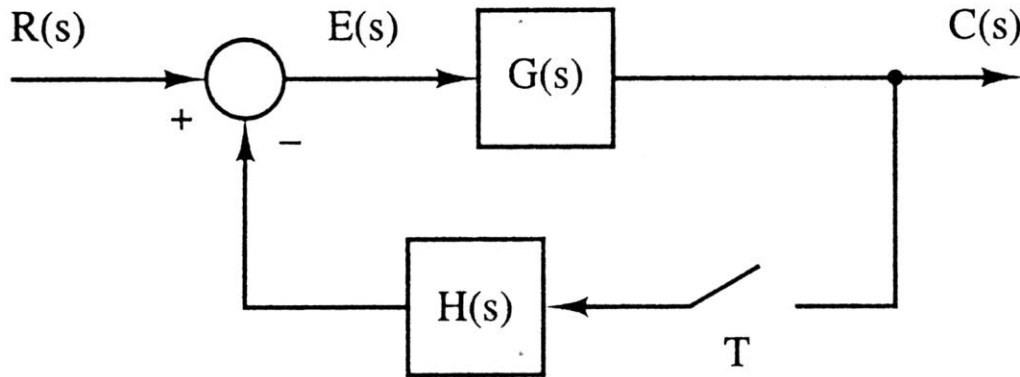


Figure 5-3 Sampled-data system.

$$C(z) = \frac{\overline{GR}(z)}{1 + \overline{GH}(z)} \quad (5-18)$$

Transfer function?

No transfer function can be derived, since $r(t)$ is not sampled.

5.3 Derivation Procedure

Finding TF for sampled-data system is difficult (due to ideal samplers).

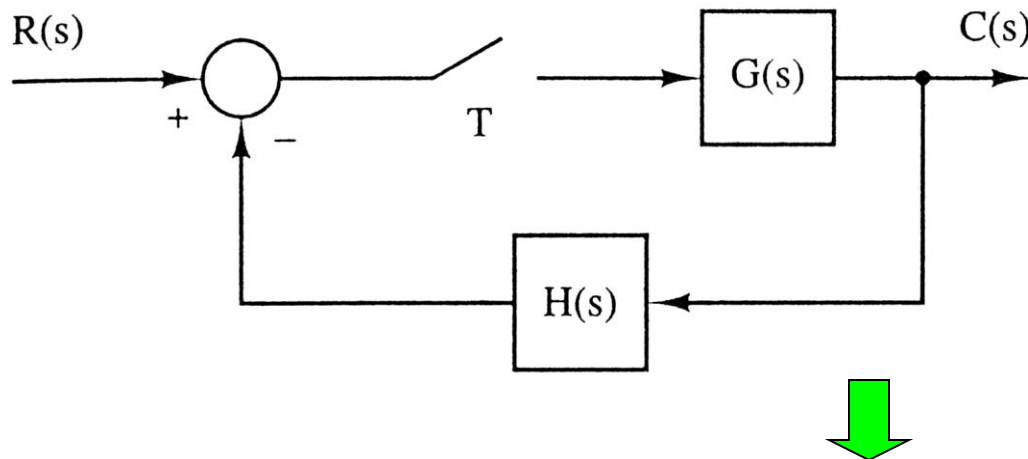
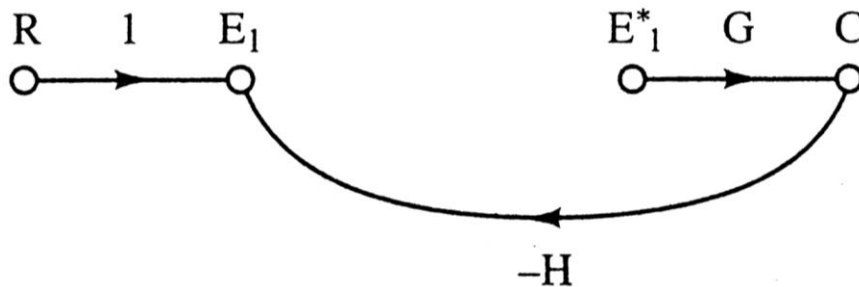


Figure 5-4 Sampled-data control system.



原始流程圖

Figure 5-5 Original flow graph for the sampled-data system in Figure 5-4.

Step-by-Step Procedure

1. Construct the *original signal flow graph* (SFG).
2. Assign a variable to each sampler input. (E_1)
3. Considering each sampler output (E_1^*) to be a source node (input), **express the sampler inputs (E_1) and the system output (C) in terms of inputs (R and E_1^*).**
4. Take the starred transform of these equations and solve by any convenient method.

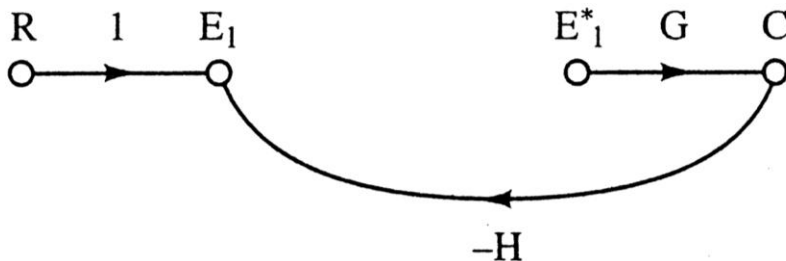


Figure 5-5 Original flow graph for the sampled-data system in Figure 5-4.

More than one sampler?

取樣流程圖

→ Cramer's rule vs. **Mason's formula** (*sampled SFG*).

Example 5.1

Consider the digital control system of Figure 5-7a which has the model given in Figure 5-7b (see Section 4.4). The original flow graph is shown in Figure 5-8. We can write

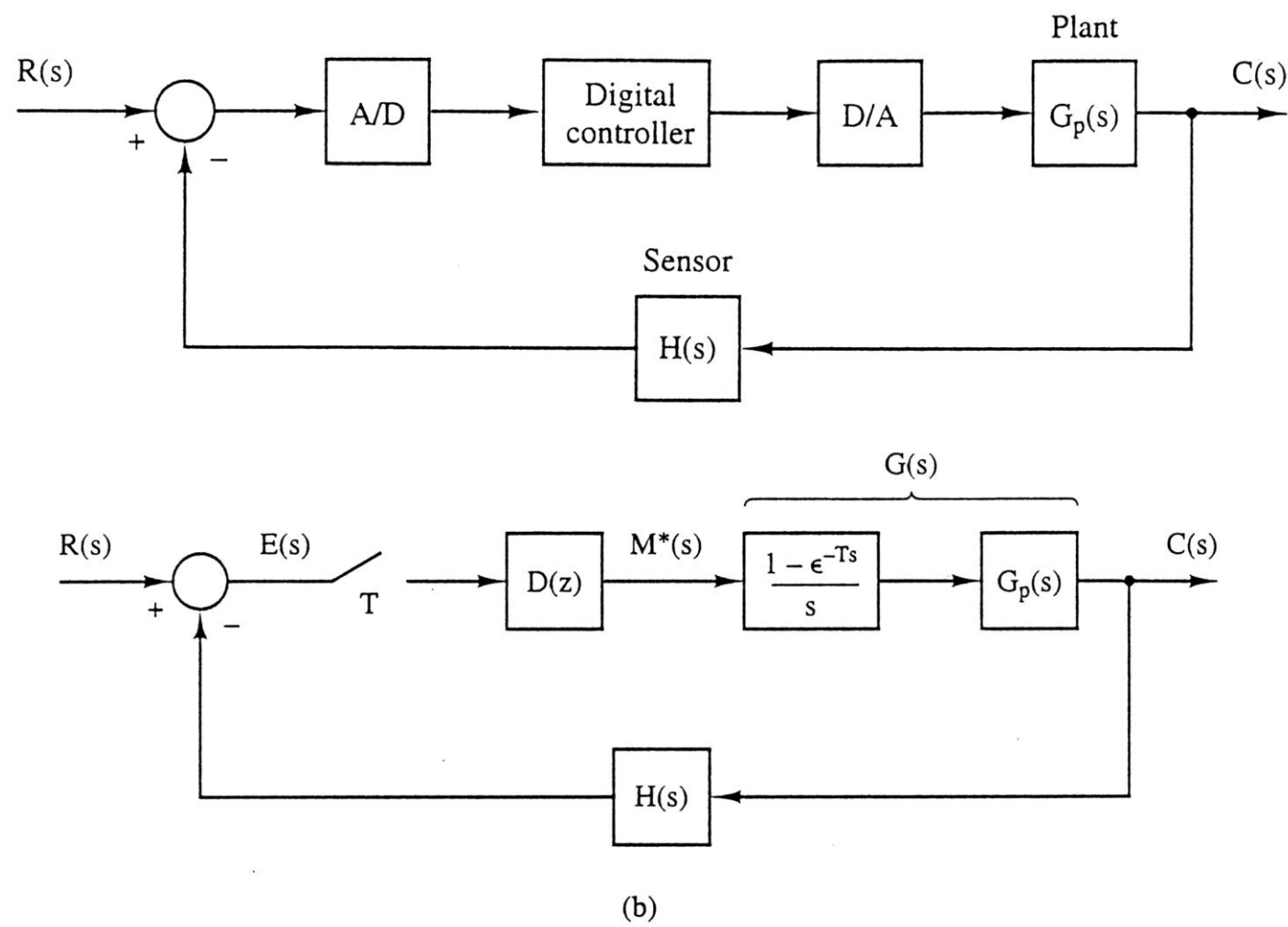


Figure 5-7 Closed-loop digital control system.

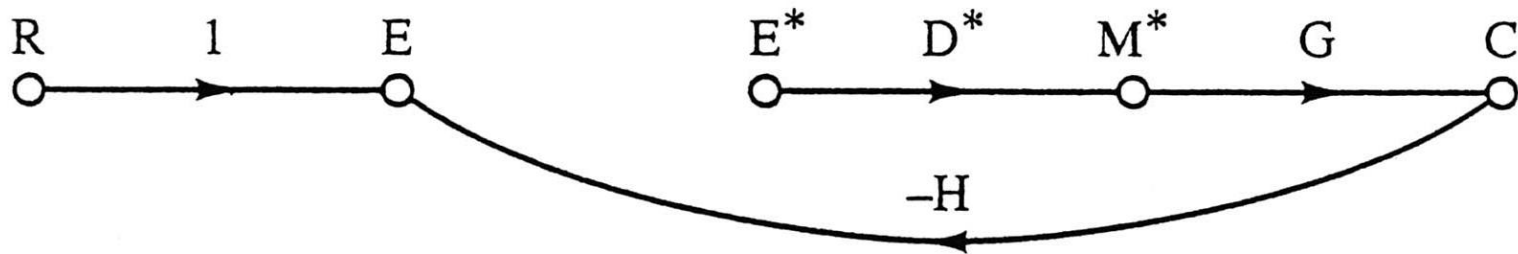


Figure 5-8 Original flow graph for Example 5.1.

$$E = R - GHD^*E^*$$

$$C = GD^*E^*$$

Hence

$$E^* = R^* - \overline{GH}^*D^*E^* \rightarrow E^* = \frac{R^*}{1 + D^*\overline{GH}^*}$$

$$C^* = G^*D^*E^*$$

Thus

$$C(z) = \frac{D(z)G(z)}{1 + D(z)\overline{GH}(z)}R(z)$$

Example 5.2

Consider the system shown in Figure 5-9. The original signal flow graph is shown in Figure 5-10. The system equations are

$$E_1 = R - G_2 E_2^*$$

$$E_2 = G_1 E_1^* - G_2 H E_2^*$$

$$C = G_2 E_2^*$$

Starring these equations, we obtain

$$E_1^* = R^* - G_2^* E_2^*$$

$$E_2^* = G_1^* E_1^* - \overline{G_2 H^*} E_2^*$$

$$C^* = G_2^* E_2^*$$

Produce Sampled SFG

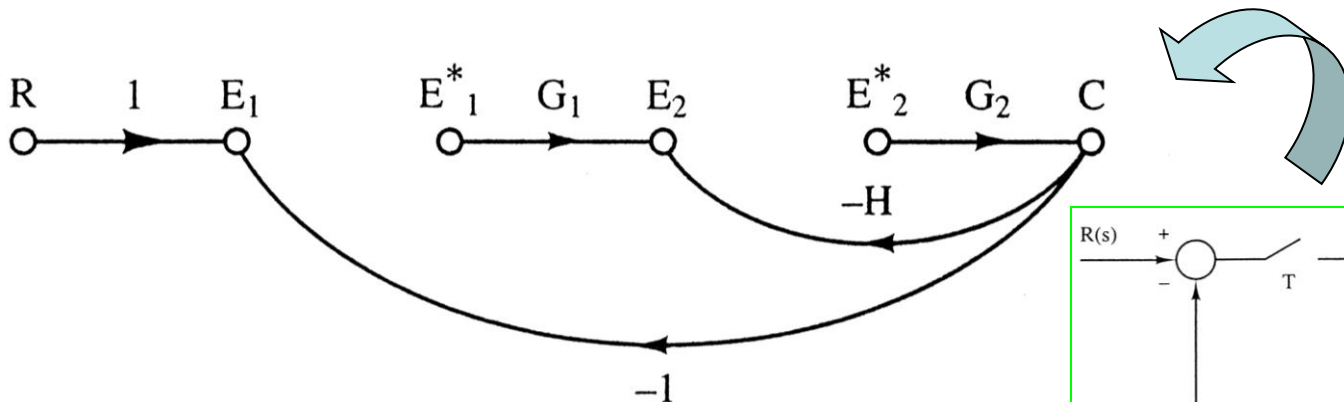


Figure 5-10 Original flow graph for Example 5.2.

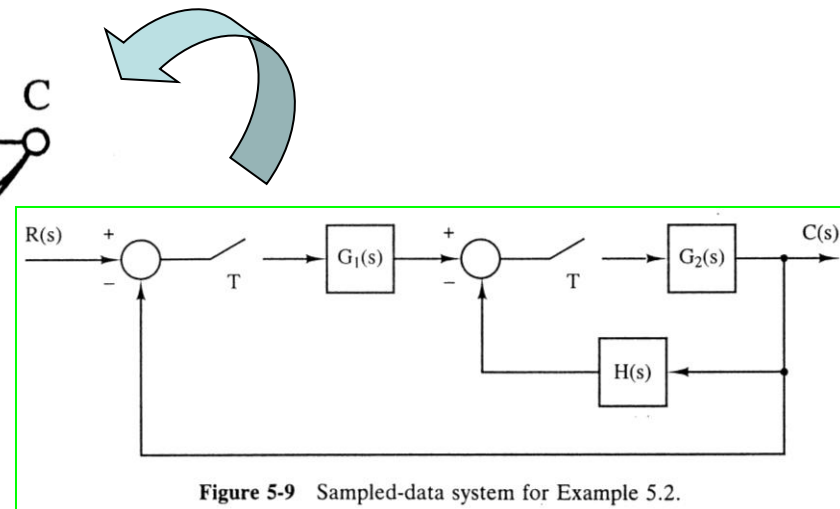


Figure 5-9 Sampled-data system for Example 5.2.

Produce Sampled SFG

The sampled flow graph can then be drawn from these equations as shown in Figure 5-11. Then applying Mason's gain formula, we obtain

$$C^* = \frac{G_1^* G_2^*}{1 + G_1^* G_2^* + \overline{G_2 H^*}} R^* \quad \text{or} \quad C(z) = \frac{G_1(z)G_2(z)R(z)}{1 + G_1(z)G_2(z) + \overline{G_2 H}(z)}$$

and

$$C(s) = \frac{G_2(s)G_1^*(s)}{1 + G_1^*(s)G_2^*(s) + \overline{G_2 H^*}(s)} R^*(s)$$

$$\begin{aligned} E_1^* &= R^* - G_2^* E_2^* \\ E_2^* &= G_1^* E_1^* - \overline{G_2 H^*} E_2^* \\ C^* &= G_2^* E_2^* \end{aligned}$$

Produce Sampled SFG

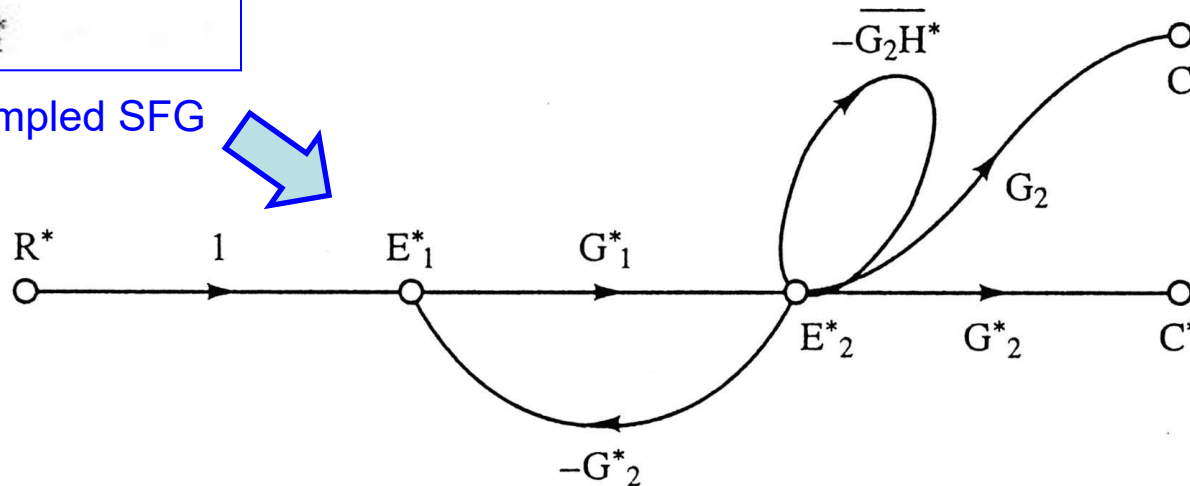
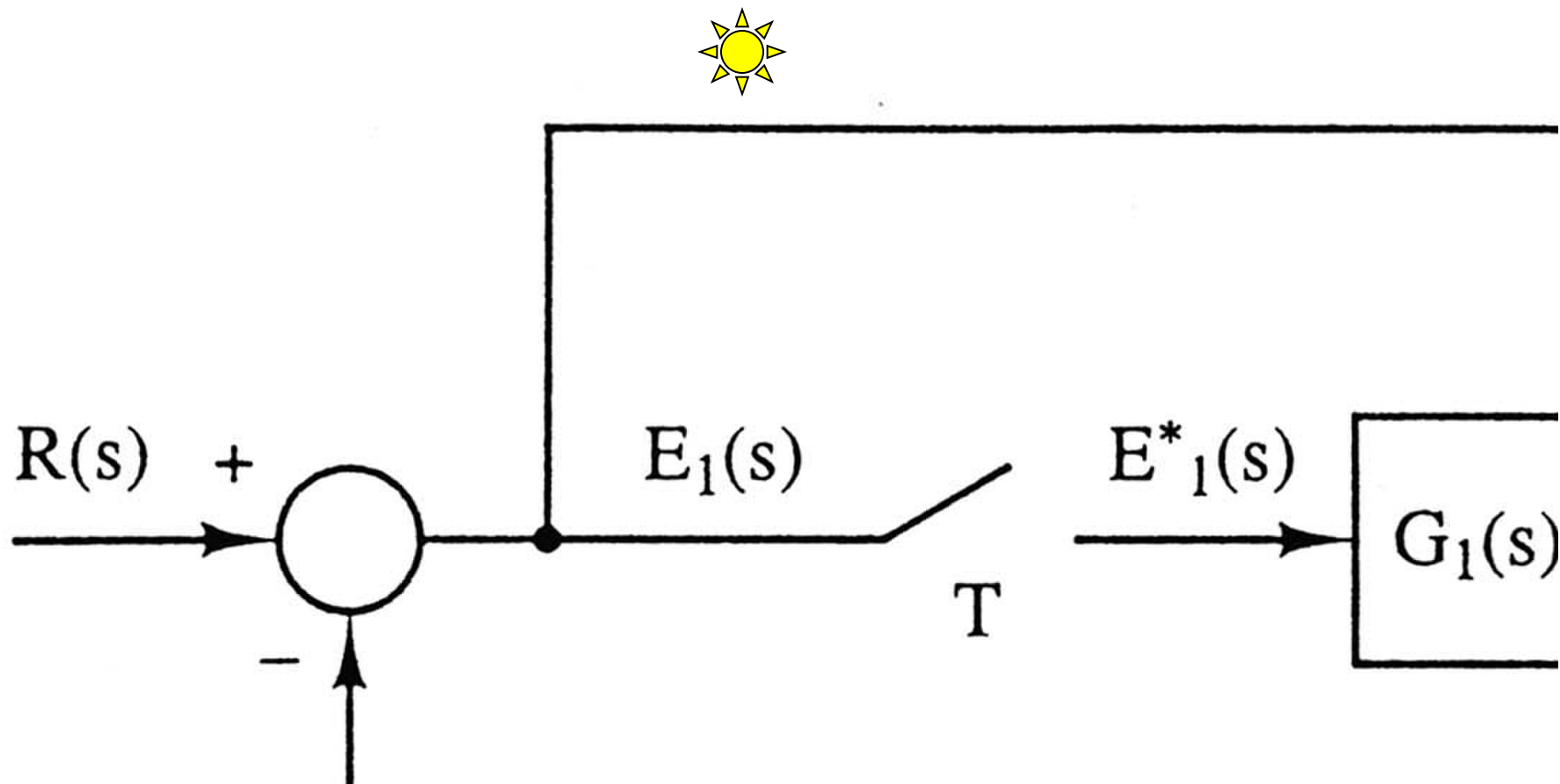


Figure 5-11 Sampled flow graph for Example 5.2.

Example 5.3

As another example, consider the system shown in Figure 5-12. Note that no transfer function may be derived for this system, since the input $R(s)$ reaches $G_2(s)$ without being sampled.

Remind: *Feedforward Matrix "D"* (Ch. 2)



The original signal flow graph is given in Figure 5-13. From this figure note that

$$E_1 = R - C$$

$$\begin{aligned} C &= E_1 + G_2[G_1 E_1^* - C] \\ &= R - C + G_1 G_2 E_1^* - G_2 C \end{aligned}$$

Therefore,

$$[1 + 1 + G_2]C = R + G_1 G_2 E_1^*$$

and

$$C = \frac{R}{2 + G_2} + \frac{G_1 G_2}{2 + G_2} E_1^*$$

$$E_1 = R - C = \frac{(1 + G_2)R}{2 + G_2} - \frac{G_1 G_2}{2 + G_2} E_1^*$$

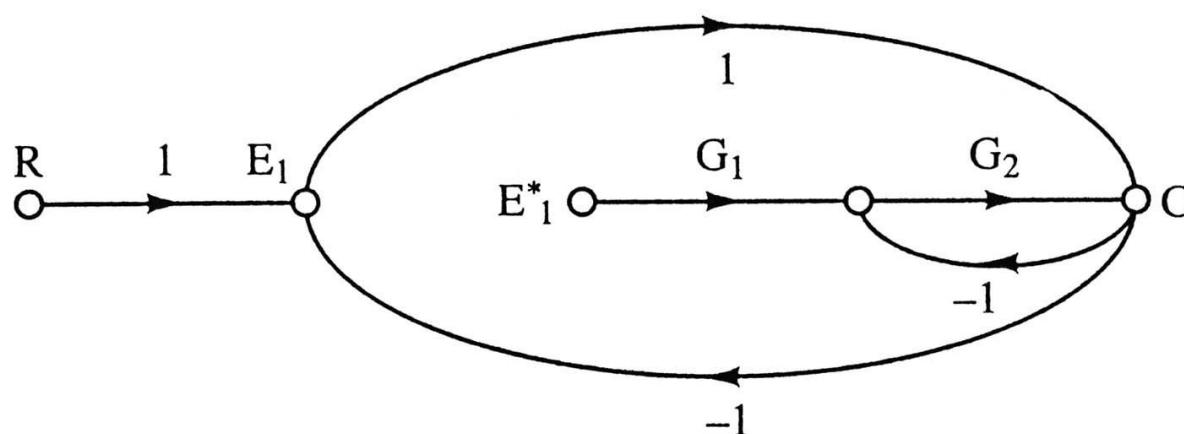


Figure 5-13 Original flow graph for Example 5.3.

Or the equation for E_1 and C can be written directly from Figure 5-13 using Mason's formula. Hence

$$C^* = \left[\frac{R}{2 + G_2} \right]^* + \left[\frac{G_1 G_2}{2 + G_2} \right]^* E_1^*$$

$$E_1^* = \left[\frac{(1 + G_2)R}{2 + G_2} \right]^* - \left[\frac{G_1 G_2}{2 + G_2} \right]^* E_1^*$$

The sampled flow graph derived from these equations is given in Figure 5-14.

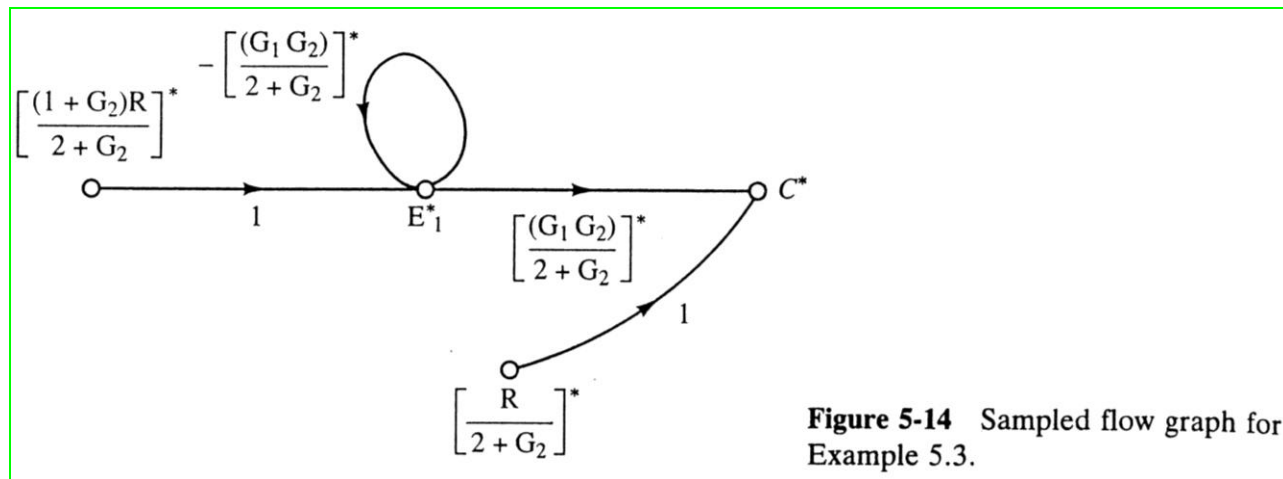


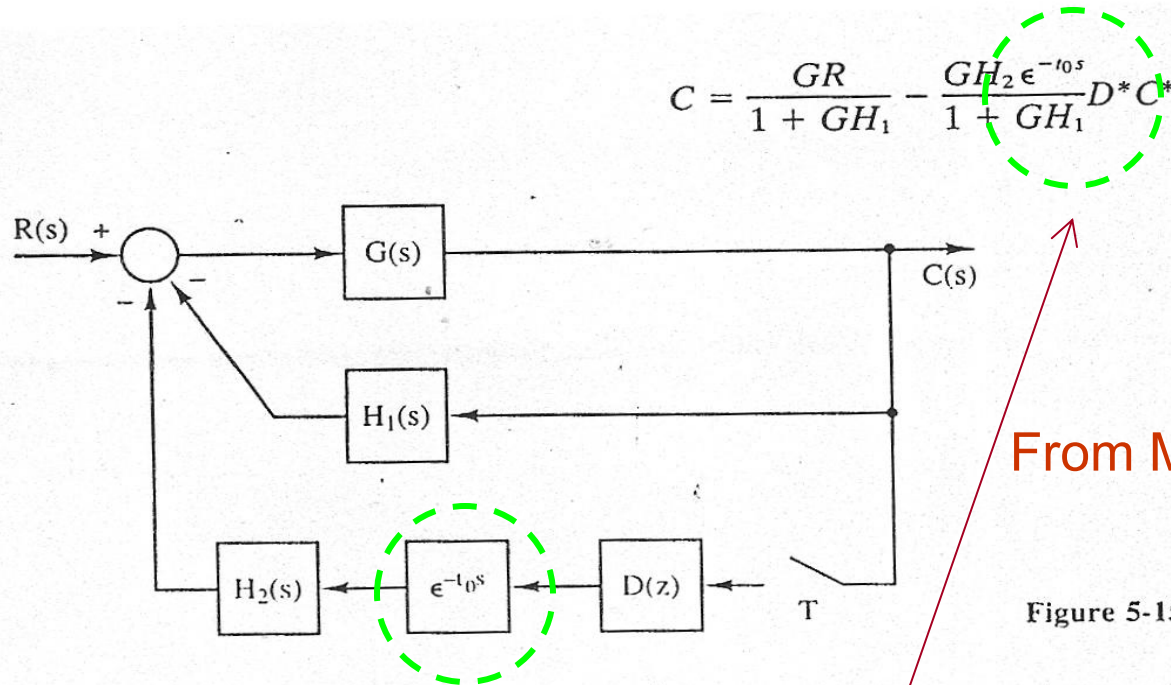
Figure 5-14 Sampled flow graph for Example 5.3.

Then by employing Mason's gain formula, we obtain

$$C^*(s) = \left[\frac{R}{2 + G_2} \right]^*(s) + \frac{\left[\frac{G_1 G_2}{2 + G_2} \right]^*(s)}{1 + \left[\frac{G_1 G_2}{2 + G_2} \right]^*(s)} \left[\frac{(1 + G_2)R}{2 + G_2} \right]^*(s)$$

Note that, as stated above, no transfer function is possible, since the input is fed into a continuous element in the system, $G_2(s)$, without first being sampled.

Example 5.4. Computation Time Delay: $t_0 < T$



From Mason's formula.

Figure 5-15 System for Example 5.4.

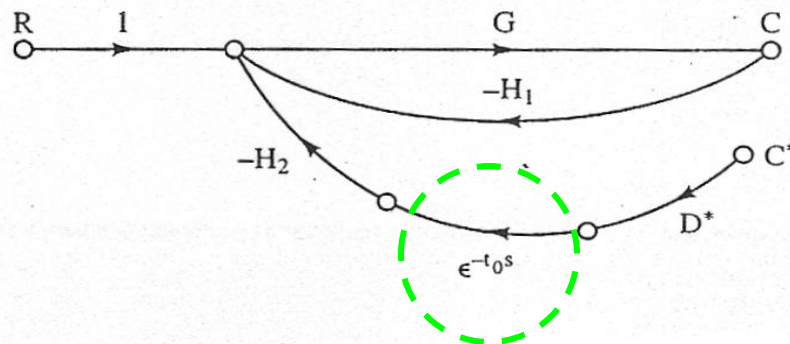
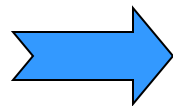


Figure 5-16 Original flow graph for Example 5.4.

- See Summary (P191)

- Problems: 1, 2, 3,
8, 9, 10, 11



Test Bank